Technical Review

2025

An Overview of Modeling Tropical Cyclone Boundary Layer (TCBL)

From Meteorological Perspectives to Wind Engineering Applications

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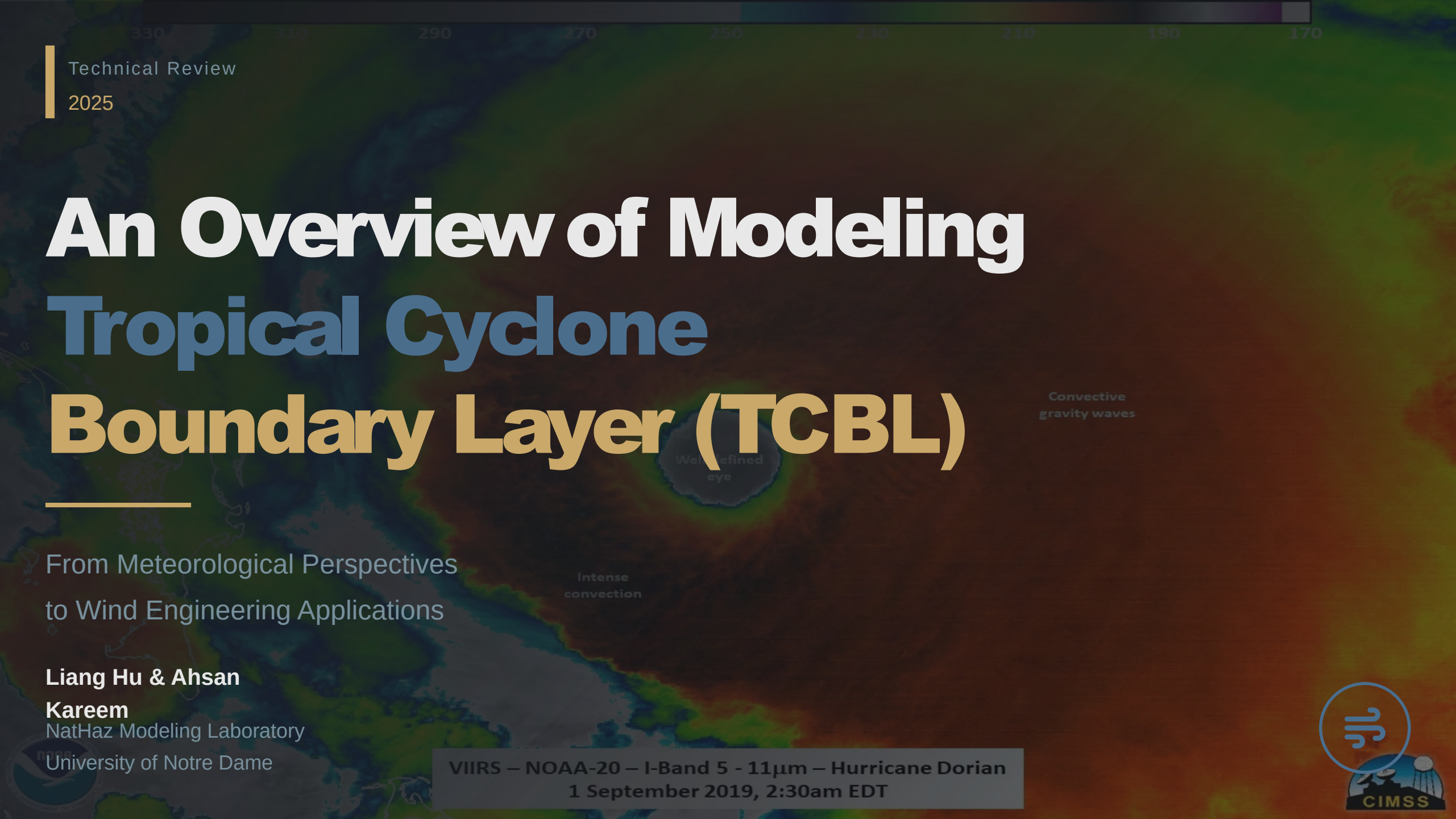

VIIRS – NOAA-20 – I-Band 5 - 11µm – Hurricane Dorian
1 September 2019, 2:30am EDT

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01

Introduction & Background

Performance-based wind engineering and the critical role of tropical cyclone boundary layer modeling

Section 1.1

Performance-Based Wind Engineering (PBWE) Framework

Framework Structure

PBWE is a **rigorous, module-based simulation framework** that evaluates the total probabilistic integral for damage prediction through Monte Carlo simulation.

Total Probability Equation:

$$G(PI) = \iiiint G(PI|DM) \cdot dG(DM|EDP) \cdot dG(EDP|IP,SP) \cdot dG(IP) \cdot dG(SP) \cdot d\lambda(IM)$$

1 Hazard Module

TC simulation providing intensity measure (wind speed & direction)

IM: Intensity Measure

2 Wind-Structure

Pressure coefficients, aerodynamic forces

IP: Interaction Parameter

3 Structural Response

Engineering demand parameters (EDP)

EDP: Engineering Demand

4 Damage & Loss

Fragility, vulnerability, performance indicators

DM: Damage Measure

Monte Carlo Simulation

1000s → **Statistical** → **Risk**

TC Events — Evaluation — Assessment

An ensemble of TC wind samples is generated based on statistics of TC characteristics, combined with sampled realizations of all parameters to yield PI samples.

Critical Challenge

Validating PBWE outcomes remains **substantially challenging** due to the absence of robust, universally accepted benchmarks.

⚠ Each framework module must be validated to the greatest extent feasible.

Key Insight: The accuracy of intensity measure from simulated TC events is essential not only for reliable PBWE results but also for establishing confidence among end users. **TCBL modeling is the critical foundation.**

Section 1.2

Multi-Scale Simulation Framework

Climate Scale
Global

Spatial
Global

Temporal
Months, Seasons, Years

Focus
Climate variability, genesis frequency, initial conditions

Macroscale
~1000 km

Spatial
~1000 km

Temporal
Several Days

Focus
TC evolution, translation, intensity change

TCBL Focus

Mesoscale
~100 km

Spatial
~100 km

Temporal
Several Hours

Focus
TCBL modeling, detailed structure

Microscale
~1-100 m

Spatial
~1 km - 100 m

Temporal
Hours - Minutes

Focus
Terrain effects, building-scale winds

TCBL's Critical Role in the Multi-Scale Framework

Central Position: The TCBL modeling operates at the mesoscale, serving as the critical link between climate/macro-scale inputs (from TC track models) and microscale engineering applications (structural wind loads).

All civil engineering structures are situated within the tropical cyclone boundary layer, experiencing the most severe wind conditions. **Accurate TCBL characterization is essential for reliable hazard assessment.**

Typical Simulation Flow

1. Generate TC events using track models
2. Apply TCBL model for 3D wind fields
3. Adjust to local gust wind speeds
4. Determine design wind speeds

Section 1.3

Tropical Cyclone Boundary Layer: Definition & Structure

Vertical Structure

The TCBL corresponds to the **lowest portion of the troposphere** where viscosity plays a significant role and cannot be neglected.

Interfacial Sublayer — ~10 cm
Molecular constraints dominate momentum exchange

Flux-Constant Layer — ~200 m
Shear-generated turbulence, constant momentum/heat fluxes

Mixed Layer — 200-1000 m
Surface influence diminishes with height

TCBL Height Definition

KW01 Definition:

Location of maximum angular momentum

Near TC center	Outside core
~200m	1200m

Typical TCBL Height: **500-1000m**

Vertical Structure Variations

With Radius
Height increases from inner to outer core

With Storm Stage
Varies during intensification, steady-state, weakening

Alternative definitions exist: by maximum total wind speed, zero inflow, or constant potential temperature

Section 1.4

Key TCBL Characteristics from Recent Research

Asymmetry

Surface wind fields exhibit notable asymmetries arising from multiple sources:

- Surface drag force and heat flux
- TC translation speed
- Vertical wind shear (moderate influence)
- Extratropical transition and Beta-gyre

Uhlhorn (2014): Asymmetries influenced by both storm motion and vertical wind shear

Inflow Angle

Wind direction exhibits significant asymmetries that vary with TC intensity and translation:

Zhang & Uhlhorn (2012):

Analyzed dropsonde measurements from **18 hurricanes** and developed parametric model for inflow angle as function of TC intensity and translation.

Low-Level Jet (LLJ)

A salient property in mean wind speed profile with maximum wind speed in TCBL:

Typical LLJ Height	Supergradient Coefficient
~500m	1.2

Successfully reproduced by Kepert (2001) and KW01. Confirmed by Franklin et al. (2003) using **600+ dropsonde profiles**.

Turbulence Structure

Vertical diffusivity and turbulence properties critical for closure models:

- Zhang & Drennan (2012): Vertical diffusivity increases from surface to peak then decreases with height
- CBLAST Program: Standard deviations and spectral densities under marine conditions
- K-profile and Louis-type closure schemes represent these characteristics reasonably well

Spatiotemporal Variations

Ahern et al. (2019): Analyzed **2,000+ dropsonde profiles** revealing systematic variations from outer to inner core and transitions across convective, post-

Measurement Advances:

Section 1.5

TCBL Model Categorization Framework

Physics vs Data-Driven
- Physics-based
- Data-driven (ML)

Full vs Diagnostic
- Full-physics (NWP)
- Diagnostic

Solution Approach
- Empirical
- Parametric
- Simulation-based

Implementation
- Solution-pursued
- No explicit solution

Linearity
- Linear
- Nonlinear

Symmetry
- Axisymmetric
- Asymmetric

Dimensionality

2D	3D
Slab / Hybrid	**Height-resolved**

7+
Categorization Dimensions

20+
Model Representations

Model Hierarchy Table

Category	Axisymmetric	Asymmetric
Diagnostic – Parametric (Empirical)	–	*Powell (1980)*
Diagnostic – Parametric (Analytical)	*Kuo (1982)*	*Kepert (2001)*
Diagnostic – Hybrid	*Smith (1968)*	*Langousis et al. (2009)*
Diagnostic – 2D–Slab	*Smith (2003)*	*Chow (1971), Vickery et al. (2000)*
Diagnostic – 3D Nonlinear	*Eliassen (1977)*	Kepert & Wang (2001)
Full–Physics (Dedicated)	*Rosenthal (1970)*	*Anthes (1971, 1981)*
Full–Physics (NWP)	–	*WRF (ARW, HWRF, HWCM, HAFS), GFS, CM1*

Hierarchy Direction
Simplified → Rigorous

Most Rigorous Diagnostic
KW01

02

Full-Physics TCBL Models

Numerical weather prediction systems and their tropical cyclone boundary layer components

Section 2.1.1

WRF-ARW: Advanced Research WRF

Model Overview

WRF-ARW (Advanced Research WRF) is the **basic research mode** of the Weather Research and Forecasting model, widely used for TC simulation and TCBL studies.

Primary Applications: TC intensity, track, structural features, and related characteristics

Key Configuration Factors

1. **Bottom Boundary**
   High-resolution surface models required. Surface exchange coefficients highly sensitive for TC intensity and TCBL structure (Ming & Zhang, 2016)

2. **Initial Conditions**
   Relies on global models (NCEP, GFS, ECMWF). Influence on TCBL during first 48 hours generally negligible (Singh & Bhaskaran, 2018)

3. **Data Assimilation**
   Local assimilation improves accuracy but generally not feasible for massive TC simulation in PBWE

4. **Ensemble Prediction**
   Multiple model configurations or perturbed initial conditions to quantify and reduce uncertainty

Turbulence Parameterizations

Two primary schemes compared extensively:

MYJ	YSU
Mellor-Yamada-Janjić	Yonsei University
Stronger LLJ	**Generally better**

Performance Evidence:

- ✓ Hu et al. (2010): YSU generally outperformed MYJ
- ✓ Nolan et al. (2009): 1.33km resolution most skillful
- ✓ Bay of Bengal studies: Larger domains benefit prediction

Overall Finding

The body of evidence suggests that **YSU generally outperforms** other commonly used parameterization schemes in ARW-based TC simulations.

Evaluation Approach

Hindcasting with model settings and parameterizations varied in sensitivity experiments

HWRF and HWCM Models

HWRF

Hurricane WRF

Specialized configuration of WRF for hurricane forecasting, continuously updated under the **Hurricane Forecast Improvement Project (HFIP)**.

Key Improvements

- Enhanced vertical diffusivity representation (Zhang et al., 2015; Zhang & Rogers, 2019)
- Continuous vertical diffusivity profile for surface wind and inflow-angle predictions (Wang et al., 2018)
- Intensified vertical mixing length effects on TCBL structure (Zhang et al., 2017b)

Status: Retired from operational forecasting, replaced by HAFS (Hurricane Analysis and Forecast System)

HWCM

Hybrid WRF Cyclone Model

Recently developed WRF mode for simulating **idealized landfalling TCs** (Bruyère et al., 2019).

Simulation Procedure

1. Generate idealized TC with randomly specified parameters
2. Allow to spin and evolve within idealized background environment
3. Insert into real environmental field to simulate landfall

Sensitivity Factors

Background conditions (SST, vertical wind shear) influence simulated TC motion and guide configuration selection

HWRF Limitations for PBWE

Computational Demand: NMM (non-hydrostatic mesoscale model) dynamical core is computationally demanding

Scalability: May limit applicability in PBWE where large numbers of TC simulations are required

HWCM Promise for PBWE

HWCM can generate TCs with randomly specified parameters while preserving full-physics representation, holding promise for future PBWE applications.

Ramos-Valle et al. (2020): Simulated ensemble of 200 TCs to analyze landfall angle effects on storm surge

Both HWRF and HWCM demonstrate that full-physics models can simulate TCBL with considerable accuracy, but **additional verification and performance enhancement are needed** before full adoption in PBWE.

Section 2.2

Other Full-Physics Models & LES Approaches

Japan Meteorological Agency

Kanada et al. (2012)

Deardorff–Blackadar nonlocal scheme performs favorably in predicting TC intensity and TCBL structure

fvGFS & RAMS

fvGFS (Hazelton et al., 2018)

FV3 finite-volume dynamical core with GFS physics. Reasonably predicts RMW and rapid intensification but struggles with TCBL height

RAMS (Abarca et al., 2015)

Luis parameterization recommended for TC simulation

Large Eddy Simulation

WRF-LES (Zhu, 2008; Huang et al., 2018)

Micro-scale resolution (~100m) for turbulence properties

CM1 (Kepert et al., 2016)

Cloud Model 1 for studying TCBL without parameterized closures

High computational cost limits PBWE application

Comparative Analysis: Full-Physics Models

Model	Dynamical Core	Key Strengths	Limitations
WRF-ARW	*Finite-difference*	*Flexible, extensively validated*	*Sensitive to parameterizations*
HWRF	*NMM*	*TC-specific improvements*	*Computationally demanding*
fvGFS	*FV3 finite-volume*	*Good RMW prediction*	*TCBL height accuracy*
CM1/WRF-LES	*LES*	*No parameterized closures*	*Extremely high cost*
JMA Model	*Non-hydrostatic*	*Nonlocal schemes*	*Limited global use*

5+

Multiple

High

03

Diagnostic Models: Foundation

Simplified models for efficient tropical cyclone boundary layer simulation

Section 3

Diagnostic Models: Concept & Governing Equations

Definition & Purpose

Diagnostic models are **simplified forms of full-physics models** that partially remove physics and do not explicitly track transient evolution. They pursue quasi-steady solutions at selected time intervals (e.g., 6 hours).

Key Characteristics

- ✓ Cover only lowest 1-3 km (TCBL height)
- ✓ Upper atmosphere represented by prescribed forcing
- ✓ One-way interaction: atmosphere → TCBL
- ✓ Computationally efficient for large-scale simulations

Model Classification

Simulation-Based

Solve equations numerically (finite difference)

Parametric

Closed-form analytical solutions

Empirical

No explicit solution process

Governing Equations

Primitive equations in cylindrical coordinates (r, θ, z):

Radial Momentum

$$u(\partial u/\partial r) + (v/r)(\partial u/\partial \theta) + w(\partial u/\partial z) - v^2/r - fv = -\partial p/\partial r + \partial/\partial z(K\cdot\partial u/\partial z)$$

Azimuthal Momentum

$$u(\partial v/\partial r) + (v/r)(\partial v/\partial \theta) + w(\partial v/\partial z) + uv/r + fu = \partial/\partial z(K\cdot\partial v/\partial z)$$

u, v, w **Wind components**	K **Vertical diffusivity**
f **Coriolis parameter**	vg **Gradient wind speed**

Simplification Hierarchy

1. Full-physics (NWP)
2. 3D Nonlinear (KW01)
3. 2D-Slab / Hybrid
4. Parametric / Linear

🔑 **Key Equation:** Given gradient wind vg as forcing at TCBL top, solve 3D wind field (u, v, w) within TCBL

Section 3.1

Axisymmetric Models: Hybrid & Height-Averaged

Axisymmetric Hybrid Models

Combine parametrized radial and vertical profiles with simulation-based methods. Numerical methods solve simplified differential equations.

Smith (1968)

Assumed wind profiles: $u = us(r)\cdot U(z)$, $v = vs(r)\cdot V(z)$. Ekman profile for U(z), V(z); solved ODEs by Runge-Kutta

Kepert (2010b)

Alternative parameterization avoiding numerical instability

Eliassen (1971)

Power series expansion with analytical solution

Foster (2009)

Separable wind profiles solved by collocation method

Extension to Asymmetric

Langousis et al. (2009): Extended Smith (1968) hybrid model to account for asymmetries by introducing azimuthally varying parameterization.

Height-Averaged Models

Integrate axisymmetric primitive equations over entire TCBL height to obtain equations for averaged wind components.

Height-Averaged Equation

$$\partial(h\cdot\bar{u})/\partial r = wh + (K\cdot\partial u/\partial z)h - CD|Vs|us$$

Where h = TCBL height, wh = vertical velocity at top

Smith (2003)

Prescribed TCBL height, two unknowns solved numerically

Smith & Montgomery (2008)

Further simplifications: linear, semi-nonlinear, geostrophic approximations

Key Characteristics

- **Supergradient Winds:** Height-averaged models reveal characteristic supergradient feature of TCBL
- **Thermodynamics:** Moisture and thermal equations can be incorporated
- **Efficiency:** Lower computational cost than full 3D models

Axisymmetric Analytical Models

Nonlinear Analytical: Kuo (1982)

Converted axisymmetric primitive equations into single equation using **complex variable approach**.

Complex Variable Formulation

$W = u + iv$, $Wg = ug + ivg$

Solution Form

$u = a_1 \cdot e^{\alpha z} + a_2 \cdot e^{\beta z}$
$v = b_1 \cdot e^{\alpha z} + b_2 \cdot e^{\beta z}$

Where α, β are radially-varying parameters in terms of gradient wind speed vg

Key Results

- ✓ TCBL vertical profiles exhibit supergradient LLJ
- ✓ Similarity between analytical and simulation-based results
- ✓ Second-order terms contribute little for Rankine-type profiles

Linear Analytical: Meng et al. (1995)

Simplified axisymmetric primitive equations by **linearizing advection terms**, enabling semi-analytical solution without momentum integration.

Solution Form

$u = a_1 \cdot e^{\alpha z} + a_2 \cdot e^{\beta z}$
$v = b_1 \cdot e^{\alpha z} + b_2 \cdot e^{\beta z}$

Introduced power series expansions for first-order (linear) and second-order (nonlinear) analytical solutions

Iterative Solution

Unknowns defined in terms of surface wind speed, requiring iterative procedure

Asymmetric Application

Can represent asymmetric TCBL by assuming asymmetric gradient pressure profile (recast as column model)

Model Relationships

- Vogl & Smith (2009): Linear analytical solution = first-order solution of Kuo (1982)
- Snaiki & Wu (2017a): Same solution reinvented as "linear weak-friction approximation"
- Kepert (2001): Axisymmetric model represents only wavenumber-0 part of 3D linear asymmetric model

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Parametric Models

Empirical and analytical solution-based approaches for TCBL wind field estimation

Section 3.2.1

Empirical Methods: Gradient Pressure/Wind Profiles

Axisymmetric Gradient Profiles

Gradient wind/pressure profiles are fundamental input to all diagnostic models. Gradient balance relates wind and pressure:

$$vg^2/r + f \cdot vg = (1/\rho) \cdot \partial p/\partial r = (\Delta p/\rho) \cdot (B/A_B) \cdot r_{B-1} \cdot \exp(-(r/A)_B)$$

Holland (1980)

$p(r) = pc + \Delta p \cdot \exp(-(r/A)_B)$

Highly adaptable with A and B parameters. Suitable for Monte Carlo simulation in PBWE. Enhanced by Holland et al. (2010) for secondary wind maximum.

Willoughby et al. (2004, 2006)

Piecewise profile with transition region. Fits nearly all measured radial TC wind profiles. More parameters make it better for meteorology than PBWE.

Asymmetric Gradient Profiles

Xie et al. (2006)

Azimuthally-varying RMW: $RMW(\theta) = RMW_0 \cdot (1 + \Sigma c_n \cdot \cos(n\theta) + \Sigma s_n \cdot \sin(n\theta))$

Polynomial coefficients determined by fitting to gale radii at four quadrants

Georgiou (1985), Tryggvason et al. (1976)

Asymmetric gradient wind from circular pressure vortex with balance along streamlines rather than isobars

⚠ **Challenge:** Limited reliable measurements for validation. Axisymmetric gradient balance validated in

TC Track Models

Emanuel (2004)

Seamlessly integrated with TC potential intensity theory. Suitable for intensity prediction.

Vickery et al. (2000b)

Auto-regression-like statistical model with parameters (translation velocity, heading, intensity) calibrated from historical records

Emanuel et al. (2006)

Statistical-physical model solving differential equation based on environmental wind and beta drift

Key Parameters

Holland-B	RMW
B	**A**
Shape parameter	Radius to Max Wind

Δp	vg
Δp	**vg**
Central Pressure Deficit	Gradient Wind Speed

Section 3.2.2

Vertical Wind Profile Models

Surface Wind Reduction Factor (SWRF)

Most straightforward approach for converting gradient wind to surface wind:

$$V_{surface} = SWRF \times V_{gradient}$$

Vickery et al. (2000a)	Batts et al. (1980)
0.865	**0.80**

Powell (1980) Evaluation

Reviewed SWRF against empirical TCBL profiles. **Simple SWRF of 0.8 outperformed complex formulations** in agreement with surface observations.

Variable SWRF

- Georgiou (1985): SWRF varies with radius and surface condition (land/sea)
- Meng et al. (1997), Ishihara et al. (2005): Parameters as functions of terrain roughness, gradient wind, and inertial stability

TCBL Profile Models

Depict vertical distribution of wind speed within TCBL, collaborating with surface wind from 2D models:

Thompson & Cardone (1996)

Distinct formulations for radial and azimuthal components with potential temperature effects. Revised by Wu & Huang (2019) with nonlinear fitting for friction velocity.

Vickery et al. (2009b)

Logarithmic-type profile inspired by linear TCBL model (Kepert, 2001):

$V(z) = (u^*/\kappa)\cdot\ln(z/z_0)\cdot f(z/h)$

Reduced RMS error by ~27% compared to constant SWRF of 0.85. Performance not exceeded to date.

Power-Law Profiles

- Meng et al. (1997), Ishihara et al. (2005): Parameters as functions of local terrain roughness, gradient wind, and inertial stability, all of which vary radially and azimuthally.

Not extensively validated against sufficiently large measurement dataset.

Profile Performance

Best Performance

Vickery et al. (2009b)

27% RMS error

Trade-off

Complex vs. simple SWRF depends on application needs

Linear Analytical Models: K01 and Column Model

Kepert (2001): K01 Model

3D Linear TCBL Model

First **3D linear TCBL model** based directly on governing equations with linearized slip boundary conditions.

Solution Form (Fourier Series)

$u = u_0(r,z) + u_1(r,z)\cdot\cos(\theta) + v_1(r,z)\cdot\sin(\theta)$
$v = v_0(r,z) + v_1(r,z)\cdot\cos(\theta) - u_1(r,z)\cdot\sin(\theta)$

Four Key Assumptions

- ✓ Vertical turbulence diffusivity K constant vertically and azimuthally
- ✓ Surface drag coefficient constant azimuthally
- ✓ Translation velocity << maximum TC wind speed
- ✓ Vertical advection ignored

Advantage: Resolves vertical structure without prescribed parameterizations, reproduces LLJ more effectively than hybrid models

Meng et al. (1995)

Column Linear Model

Axisymmetric model reformulated as **column model** varying azimuthally by introducing asymmetric gradient balance.

Asymmetric Gradient Wind

$vg(r,\theta) = vg0(r) + vg1(r)\cdot\cos(\theta) + \ldots$

Key Feature: Solutions at different azimuths or radii are independent, making formulation effectively a set of column models indexed by azimuth and radius

Difference from K01: Asymmetric gradient-wind approach cannot fully capture height-resolved azimuthal advection present in K01 solution

Recent Developments

Li et al. (2020)

Introduced height-dependent turbulence diffusivity into K01 model by solving additional ODEs iteratively using central difference method

Limitation: Computational cost not substantially lower than fully nonlinear model

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Simulation-Based Models

2D and 3D numerical approaches for TCBL wind field simulation

Section 3.4

2D-Slab Model: Formulation & Applications

Model Formulation

Classical 2D-slab TC model on **height-averaged primitive equations** in moving Cartesian coordinates (Vickery et al., 2000a).

Governing Equations

$\partial U/\partial t = F + A + P + E + D$

$U = (\bar{u}, \bar{v})$: vertically-averaged wind components
F: Coriolis, A: Advection, P: Pressure, E: Diffusion, D: Surface Drag

Key Limitations

- Does not resolve vertical structure
- Cannot explicitly represent vertical advection or diffusivity
- TCBL height must be specified a priori (typically 1000m)
- Cannot produce height-resolved TCBL wind field

Numerical Solution

Chow (1971)
First-order upwind finite difference scheme with nested grid system (7 layers, 2km innermost spacing)

Grid System
Staggered grid arrangement for evaluating central differences, full algorithmic details provided

Exploring TC Physics

Shapiro (1983)
Investigated translation-induced asymmetry. Nonlinear effects minor at VT < 5 m/s but substantial at 10 m/s.

Williams (2015)
Momentum budget analysis relative to 3D nonlinear model. Highlighted critical importance of nonlinear asymmetric advection.

⚠ **Limitation:** Height-averaging limits ability to characterize complex TC physics

Surrogates for Efficiency

Shapiro (1983) - Intrusive
Spectral-method-like approximation with Fourier series. Solve only Fourier coefficients via differential equations.

Vickery et al. (2000a) - Non-intrusive
Apply Fourier series to Chow's algorithm solutions. Linear interpolation for coefficients reduces storage and computation.

PBWE Applications

- HAZUS: FEMA's hazard assessment
tool

Section 3.5

2D Hybrid Models

Langousis et al. (2009)

Extended Smith (1968) hybrid axisymmetric model to **asymmetric formulation** by introducing azimuthally varying parameterization.

Wind Profile Parameterization

$u(r,\theta,z) = us(r,\theta)\cdot U(z; r,\theta)$
$v(r,\theta,z) = vs(r,\theta)\cdot V(z; r,\theta)$

Solution Method

Unknowns us, vs solved by substituting into nonlinear primitive equations and height-averaging. Nonlinear iteration combined with central difference method. Computational cost comparable to 2D-slab models.

Theoretical Advantages

- **All Nonlinear Terms Retained:** Including vertical advection
- **Should Outperform 2D-Slab:** From theoretical perspective

Limitations

Assumption Constraints

- Constant vertical turbulence diffusivity K
- Linearized surface drag
- Limits ability to capture key nonlinear features

LLJ Prediction

Expected lower accuracy in predicting LLJ characteristics compared to KW01 model

Validation Issues

KW01 not used as benchmark in original study. Some comparison model results not always reliable.

Potential Improvements

- Adopt vertically varying K profiles (similar to Bode & Smith, 1975 enhancements)
- Rigorous correction and calibration against established models and measurements

Future Potential

If properly corrected and calibrated, this formulation has potential to deliver meaningful benefits for PBWE applications by combining computational efficiency with theoretical rigor.

Position in Hierarchy: 2D hybrid models lie between 2D slab and 2D models. Validation

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3D Nonlinear Models (KW01)

The most rigorous diagnostic TCBL formulation and benchmark for wind engineering

Section 3.6.1

KW01 Model: Development & Formulation

Historical Development

Anthes (1972, 1981)

Early asymmetric model with vertical/horizontal diffusion and air-sea exchange. Only 3 vertical levels (1 in TCBL).

Subsequent Enhancements

Boundary-layer parameterization, nested grids, idealized profiles, data assimilation, verification (Elsberry, 1979; Anthes, 1982)

Kepert & Wang (2001)

Most rigorous diagnostic TCBL formulation to date. Targeted simplifications for TCBL-focused studies.

Targeted Simplifications

- ✓ Steady-state solutions
- ✓ Prescribed idealized gradient pressure
- ✓ Removed mass variation effects
- ✓ Dry atmosphere assumption
- ✓ Hydrostatic approximation

ℹ Appropriate when TCBL characteristics are primary focus

Numerical Methods

Time-Splitting Strategy

Each time step divided into advection, adjustment, and physics stages

Spatial Discretization

Third-order upwinding scheme (Wang, 1995, 1996) with forward-backward explicit-implicit algorithm

Coordinate System

Terrain-following, TC-center-following coordinates with asymmetry in surface drag from TC translation

Turbulence Closure

MYJ level-2.25 (depends on TKE and Richardson number). Other schemes can be substituted.

Physics Included

✓ Energy equation: Effect of sensible heat on density under hydrostatic assumption

Excluded:

Humidity (minor influence in TCBL context), cloud microphysics, air-sea coupling, deep convection

Key Successes

🏆 Successfully reproduces key TCBL features including LLJ, effects of vertical advection and

KW01 Validation & Underlying Physics

Validation Studies

Kepert et al. Series (2002-2009)

Comprehensive validation against dropsonde data:

- Hurricane Georges (1998)
- Hurricane Mitch (1998)
- Hurricane Danielle (1998)
- Hurricane Isabel (2003)

Calibration Procedure

Objective analysis of TC track, fitting flight-level data to minimize RMS error in radial gradient pressure profile

Additional Validations

Powell et al. (2009), Hong et al. (2019), Zieger et al. (2021), Done et al. (2020)

Critical Validation Issues

- **Indicators:** Need appropriate metrics to quantify accuracy and uncertainty systematically
- **Ensemble Assessment:** Performance should be evaluated probabilistically in ensemble sense

Underlying Physics Investigations

Turbulence Closure (Kepert, 2012)

Evaluated 4 schemes (Bulk-Hi-Res, Louis, MYJ, KPP). **Louis and MYJ recommended** for lower RMS errors.

Vertical Advection (Fei et al., 2021)

Vertical advection of radial wind increases LLJ height/strength by **~50%**. Advection of azimuthal wind increases height by ~30% but decreases strength by ~10%.

Landfall (Kepert, 2002; Williams, 2019)

Key features: wind speed decay, secondary surface wind maxima, discontinuities in surface isotachs, landfall-motion asymmetry analogy

Thermodynamics (Williams, 2016, 2017)

Extended KW01 with humidity, cloud water, rainwater equations for inner-core thermodynamics and eyewall replacement

TC Evolution & Micro-Scale

- **Eyewall Replacement (Kepert, 2013):** Concentric eyewalls represented through parametric profiles
- **Roll Vortices (Gao & Ginis, 2014):** Linear perturbation equations derived from axisymmetric KW01 basic state
- **Rainbands (Kepert, 2018):** Modified with nonlinear balance equation for prescribed spiral vorticity structure

KW01 Comparisons & Applications

Comparison to 2D-Slab Models

Stationary TCs

- 2D-slab deviates more from gradient wind
- Excessive inflow (larger inflow angles)
- Unrealistic sensitivity to Coriolis parameter
- Artificial oscillations when drag reduced

Moving TCs

- Translation-induced asymmetry several times larger in 2D-slab
- Differences from surface friction AND vertical adjustment
- Excessive oscillations suppressed by empirical adjustments

> "The tuning of adjustments has concealed fundamental deficiencies in the model." — Kepert (2010a)

Comparison to Linear Models

Four Key Differences: Nonlinear terms, turbulence closure, vertical advection, spatially varying drag
LLJ Strength: Nonlinear model captures peak jet near eyewall from inertial instability (linear fails)
Sensitivity: KW01 surface wind less sensitive to small gradient pressure perturbations

Comparison to Full-Physics Models

CM1 (Fei et al., 2021)

<5% relative difference in LLJ strength and height. Broader TCBL structures show noticeable differences.

WRF-ARW (Ramsay et al., 2009; Kepert & Nolan, 2014)

KW01 captures most significant TCBL features. Linear approximation sometimes agrees more closely with WRF than nonlinear KW01.

Stern (2010)

Good agreement with KW01 in LLJ height and intensity from idealized WRF simulations

PBWE Applications

Current Limitations

- No open-source or commercial implementations
- Accuracy not fully established for PBWE-relevant characteristics
- Computational demand higher than parametric models
- Little practical incentive to transition from established models

Done et al. (2020)

Time-varying diagnostic version applied to **714 historical global TC footprints**. Required ~216 CPU hours for 24 hours of TC evolution.

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Data-Driven Methods & NatHaz Work

Machine learning approaches and systematic investigation at the University of Notre Dame

Data-Driven Methods & NatHaz Investigation

Data-Driven Approaches

Modifying Radial Profiles

Xie et al. (2006), Hu et al. (2012): Azimuthally varying RMW and parameters calibrated against gale radii. Match well with H*Wind but difficult to generalize.

Introducing Asymmetries

Knaff et al. (2007, 2018): CLIPER model with wavenumber-1 asymmetry. Chang et al. (2020): Superposed wavenumber-1 on Holland profile with VWS effects.

POD-Based Models

Loridan et al. (2017): Reconstruct TCBL surface wind fields using POD modes from WRF-simulated wind fields with random forest for modal coordinates.

Data-Driven Limitations

- No clear evidence one method consistently outperforms others
- Training/evaluation based on limited TC observations
- Advantages over physics-based models not adequately demonstrated

Future: Machine learning advances (e.g., GraphCast) expected to make data-driven surrogates more viable

NatHaz Investigation Overview

Systematic study at University of Notre Dame (Hu, 2025) surrounding 3D nonlinear TCBL model with four major objectives.

1. **Factor Evaluation**
   Effects of algorithm order, thermal effects, vertical diffusivity, surface drag coefficients. Influence within 22% across TC parameter range.
2. **Benchmark Dataset**
   692 sorties combining Holland profiles (Flight+) with measurements (surface stations, SFMR, H*Wind, dropsonde). 82 sorties with all three output types.
3. **Linear Model Hierarchy**
   Enhanced linear models with height-dependent diffusivity, asymmetric gradient winds, thermal profiles. Height-resolving column model as reasonable approximation.
4. **Land-Sea Roughness**
   Landfall-induced asymmetry: ~22% change in over-ocean wind speeds, ~14% in surface winds. Decomposed into transitional and global distortion components.

Key Findings

- Enhanced TCBL modeling framework improves reliability of PBWE-enabled wind-resistant design. Pathway established for incorporating 3D nonlinear model into PBWE workflows.

692	22%	4	14-22%
Benchmark Sorties	Max Factor Influence	Research Objectives	Landfall Wind Change

2025

Conclusions & Future Research Directions

Key Conclusions

1. **KW01 is the most rigorous diagnostic TCBL model** and suitable benchmark for PBWE. Models at lower simplification levels cannot match its accuracy.

2. **Full-physics models (WRF, CM1)** resolve all physical processes but remain computationally excessive for PBWE at this time.

3. **Diagnostic models form a hierarchy** from simplified (parametric, 2D-slab) to rigorous (3D nonlinear), each with trade-offs in accuracy and cost.

Future Research Needs

- **Incorporate KW01 into PBWE** with statistical uncertainty models
- **Investigate alternative radial profile models**
- **Comprehensive model comparisons** using benchmark dataset
- **Develop land-sea roughness statistical models**
- **Extend to micro-meteorological phenomena**
- **Assess model uncertainty impact** on PBWE metrics
- **Continue benchmark dataset development**
- **Integrate with machine learning approaches**

Bridging Meteorology and Wind Engineering

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